Younes Mouhib

MSc Mathematics, ETH Zurich (2026) · optimisation · combinatorics · probability · verified computation · available immediately

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French citizen · EU/EFTA free movement, Swiss B permit · no sponsorship required · born 21.08.1998
Website · GitHub · LinkedIn · arXiv · Google Scholar

PROFILE

MSc Mathematics, ETH Zurich (final grade 5.2/6, thesis 6.0/6), after a BSc at EPFL and *classes préparatoires* MP*. Sole author of the current record lower bounds for the Hales–Jewett numbers, established under a discipline in which every claim is re-derived by independent, hand-auditable code. Toolkit: optimisation (mixed-integer, convex, network), probability and stochastic processes, statistics, Python.

EDUCATION

ETH Zurich — MSc Mathematics — Sept 2023 – Aug 2026 · Zurich, Switzerland
Final grade 5.2/6 · Master's thesis 6.0/6

- Selected coursework: Network & Integer Optimization · Structural Graph Theory · Graph Theory · Algebraic Methods in Combinatorics · Algebraic Topology.
- Master's thesis (advisor: Prof. Dr. Lorenz Halbeisen), *Improving Bounds on Hales–Jewett Numbers: Symmetric Colorings, SAT Solvers, Line-Family Variants, and Forcing Structures* — an exact symmetry reduction collapsing an intractable search over $[3]^{21}$ into a polynomial-size constraint model, with solver-independent certificates.

EPFL — BSc Mathematics — Sept 2018 – Jul 2023 · Lausanne, Switzerland

- Selected coursework: Measure & Integration · Ergodic Theory · Stochastic Processes · Discrete Optimization.
- Bachelor thesis (12 ECTS, 5.5/6), advisors Prof. Dr. Maria Colombo and Dr. Lucio Galeati: *On the Subadditive Ergodic Theorem and Its Applications*.

Lycée Masséna — Classes préparatoires MPSI/MP* — 2016 – 2018 · Nice, France

PUBLICATION

Y. Mouhib. *Lower Bounds for the Hales–Jewett Numbers via Symmetric and One-Weight Colorings*. Preprint, arXiv:2606.22155 [math.CO], 2026, 46 pp.

- Record lower bounds $\mathrm{HJ}(3,3) \geq 22$ and $\mathrm{HJ}(4,2) \geq 14$ (previous best 14 and 12); certificates reproduced in seconds by dependency-free verifiers (hj-certificates, CI-tested, Zenodo DOI 10.5281/zenodo.21537624). Supersedes arXiv:2606.22155v1 (with L. Halbeisen) and arXiv:2607.02226.

RESEARCH

Semester Project — ETH Zurich (Prof. Dr. Lorenz Halbeisen) — 2025 · Zurich, Switzerland
From Hypercubes to Sparse Graphs: Structural Translation of the Hales–Jewett Theorem via Monochromatic Stars

Student Seminar — ETH Zurich (organised by Prof. Dr. Benny Sudakov) — 2025 · Zurich, Switzerland
The Graph Regularity Method

SELECTED PROJECTS

- **hales-jewett-explorer** — interactive browser companion to the thesis; every certificate re-checked on the page.
- **near-momentum-bot** — cost-aware backtesting and walk-forward validation (Python).
- **brigade** — code generation gated by an independent Lean 4 (Mathlib) proof verifier.

TEACHING & PROFESSIONAL EXPERIENCE

Course Instructor (PVK), Topology — VSETH, ETH Zurich — Jun 2026 · Zurich, Switzerland

- Four-day intensive exam-preparation course for 20+ students: general topology, the fundamental group, Seifert–van Kampen.

Homework Tutor, grades 9–11 — Collèges Villamont & St-Roch — Jul 2021 – Jan 2023 · Lausanne, Switzerland

- Mathematics and science support; individual study plans building independent working habits.

Student Mentor — EPFL — Sept 2020 – Feb 2021 · Lausanne, Switzerland

Studies self-financed throughout by part-time work alongside full-time study (logistics, hospitality, delivery, retail, tuition). Payslips on request.

INTERESTS

Literature (300+ books read in French) · classical and flamenco guitar (10+ years) · martial arts (10+ years) · Project Euler and Mathraining.

SKILLS & REFEREES

Quantitative	Probability and stochastic processes · statistics · time series · convex, network and mixed-integer optimisation · graph theory · SAT and constraint solving · walk-forward, cost-aware backtesting
Programming	Python (NumPy, pandas, Matplotlib, pytest) · SQL · MATLAB · Lean 4 · Git and CI · LaTeX · Excel
Languages	French (native) · English (fluent) · Arabic (fluent) · Spanish (conversational) · Indonesian (intermediate) · German (beginner, actively learning)
Referees	Prof. Dr. Lorenz Halbeisen (ETH Zurich) · Prof. Dr. Maria Colombo (EPFL) · Dr. Lucio Galeati (Univ. dell'Aquila, formerly EPFL) — contact details on request